

Applied Physics (NS1001)

Date: 22nd September, 2025

Course Coordinator:

Prof. Dr. Saman Shahid

Sessional-I Exam

Total Time (Hr):

Total Marks:

Total Questions:

1

25

2

Do not write below this line

- Attempt all the questions.
- Properly mention the question number and the statement on the answer sheet.
- Write answers to each question in sequence on the answer sheet.
- Take the value of $g=9.8 \text{ ms}^{-2}$

CLO # 01 (Apply vector algebra to solve force and motion in 2/3 dimensions)

Question 1 [Total Marks: 10]

Q1(a): The displacement from Miami to the hurricane is $\vec{B} = (173\text{km})\hat{i} + (-100\text{km})\hat{j}$; the displacement from the initial position of the reconnaissance aircraft to the hurricane is $\vec{C} = (173\text{km})\hat{i} + (-200\text{km})\hat{j}$. Use the dot product to determine the angle between these two vectors. Also, show the vectors graphically. [3+2=5 marks]

Q1(b): A projectile is fired horizontally from a gun that is 45.0 m above flat ground, emerging from the gun with a speed of 250 m/s. (a) How long does the projectile remain in the air? (b) At what horizontal distance from the firing point does it strike the ground? (c) What is the magnitude of the vertical component of its velocity as it strikes the ground? [1+2+2=5 marks]

CLO # 01 (Apply vector algebra to solve force and motion in 2/3 dimensions)

Question 2 [Total Marks: 15]

Q2(a): An athlete participates in a long jump competition, leaping into the air with a velocity, v_0 at an angle θ_0 With the horizontal. Show that the trajectory of the athlete forms a parabola in terms of v_0 , θ_0 , and g . [5 marks]

Q2(b): Tarzan, who weighs 820 N, swings from a cliff at the end of a 20.0 m vine that hangs from a high tree limb and initially makes an angle of 22.0° with the vertical. Assume that an x-axis extends horizontally away from the cliff edge and a y-axis extends upward. Immediately after Tarzan steps off the cliff, the tension in the vine is 760 N. Just then, what are:

(a) the force on him from the vine in unit-vector notation and the net force on him

(b) in unit-vector notation

and as (c) a magnitude

and (d) an angle relative to the Positive direction of the x-axis?

What are the

(e) magnitude

and (f) angle of Tarzan's acceleration just then? [2+2+2+2+1+1=10 marks]